

*MA.8.AR.3.4***Benchmark**

MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table or an equation in slope-intercept form.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.4.3
- MA.8.AR.4.2, MA.8.AR.4.3
- MA.8.DP.1.3

Terms from the K-12 Glossary

- Intercept
- Linear Equation
- Slope

Vertical Alignment**Previous Benchmarks**

- MA.6.GR.1.1

Next Benchmarks

- MA.912.AR.2.4

Purpose and Instructional Strategies

In previous courses, students plotted rational number ordered pairs in all four quadrants and on both axes. In grade 7 accelerated, students graph proportional relationships from a table, equation or a written description. Students also graph an equation from slope-intercept form from a written description, a table, a graph or an equation. In Algebra 1, students will graph a linear function when given a table, equation or written description.

- Point-slope form and standard forms are not expectations at this grade level.
- Review the concept of slope from MA.8.AR.3.2 for students who may need additional work to determine the slope and understand the meaning of slope.
- The instruction includes examples where the slope is positive or negative and the y-intercept is given as a positive or a negative in the equation.
- When introducing the benchmark, review graphing on the coordinate plane and determining appropriate scales for the graph.
- Instruction includes the understanding that a real-world context can be represented by a linear two-variable equation even though it only has meaning for discrete values. Discussing discrete values will prepare students to represent domain and range of real-world contexts in later courses.
 - For example, if a gym membership cost \$10.00 plus \$6.00 for each class, this can be represented as $y = 10 + 6c$. When represented on the coordinate plane, the relationship is graphed using the points (0,10), (1,16), (2,22), and so on.
- For mastery of this benchmark, students should be given flexibility to represent real-world contexts with discrete values as a line or as a set of points.

Common Misconceptions or Errors

- Students may incorrectly identify the slope and y-intercept.
- When graphing, students may incorrectly graph the line by inverting the directions of the slope values.
 - For example, if the slope is $\frac{2}{3}$, a student may think that 2 represents the change in the horizontal direction rather than the vertical direction.

Strategies to Support Tiered Instruction

- Teacher supports students who incorrectly identify the values for the slope and y-intercept by providing opportunities to notice patterns between a given value for b , a line graphed on the coordinate plane, and a given equation of the same line.
- Teacher supports students who invert the x - and y -values when calculating slope by using real-world problems that students can relate to and helping students represent the relationship visually.
- Instruction includes supporting students who incorrectly graph the line by inverting the directions of the slope values. Students may incorrectly calculate the slope with a common error of inverting the change in y and the change in x . Teachers can support students using error analysis tasks, in which the expression $\frac{y_2 - y_1}{x_2 - x_1}$ is incorrectly written as $\frac{x_2 - x_1}{y_2 - y_1}$.
- Instruction includes having students find the error and make corrections.
- Teacher supports students who incorrectly graph the slope of a given line through error analysis tasks, in which a line is incorrectly graphed by inverting the change in y and the change in x and then have students find and correct the error.
- Teacher co-creates an anchor chart naming the slope and y-intercept of a given line and then discusses where to start when graphing the line.
- Teacher provides graphs and equations of several linear equations then co-illustrates connections between the slopes and y-intercepts of each line to the corresponding parts of each equation using the same color highlights.
- Teacher co-creates a graphic organizer with students to include examples of positive and negative slope; the meaning of each variable in slope-intercept form; and how to determine the slope and y-intercept in a table, graph and verbal description.
- Teacher provides instruction on creating an equation table to clear up the misconception of incorrectly graphing an equation on a coordinate plane.

x	$y = 2x + 5$	y	Coordinate Points
0	$y = 2(0) + 5$ $y = 0 + 5$ $y = 5$	5	(0,5)
-2	$y = 2(-2) + 5$ $y = -4 + 5$ $y = 1$	1	(-2,1)
6	$y = 2(6) + 5$ $y = 12 + 5$ $y = 17$	17	(6,17)

- Teacher provides instruction on determining the slope and y-intercept when reading verbal description.

Instructional Tasks

Instructional Task 1 (MTR.6.1, MTR.7.1)

Brent wants to buy a 60" LED Smart TV. He opened a savings account and added money to the account every month. The table below shows the relationship between the number of months Brent has been saving and the total amount of money in his account.

Savings Account	
Number of Months	Total Savings
1	\$160
4	\$415
6	\$585
9	\$840
12	\$1,095

Part A. Graph the relationship on a coordinate plane.

Part B. If the new Smart TV costs \$1500 and tax will be \$110, approximately how many more months does he need to save money in order to make the purchase?

Instructional Task 2 (MTR.3.1, MTR.4.1)

Part A. Graph $y = .25x - 3.5$ on the coordinate plane.

Part B. Discuss with a partner your method of graphing the equation of the line.

Instructional Items

Instructional Item 1

Graph $y = x - 2$ on the coordinate plane.

Instructional Item 2

Supplies for the car wash cost \$25. The booster club is charging \$10 per car. Graph the relationship between the amount of money earned and the number of cars washed.

Instructional Item 3

The table shown represents a linear relationship. Use the table to graph the relationship

x	y
-1	2
0	0
1	-2
2	-4

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*